Dilemma Dynamics

Cooperation, deliberation and repeated-game incentives

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Evaluation and mathematical notes

Abstract

Does drafting before acting help an AI recognise when cooperation pays? Dilemma Dynamics studies repeated Prisoner's Dilemma games, where the immediate gain from defection competes with the future value of reciprocity. We compare immediate action with a bounded private draft in NVIDIA's Nemotron Nano and Super at two sampling temperatures. Against disclosed opponent policies, exact dynamic programming measures the payoff given up at each choice. Autonomous pairings reveal how cooperation begins, breaks down and returns. Drafting lowers average benchmark loss for both models at both temperatures, yet increases loss against Tit-for-Tat. Autonomous self-play also becomes less cooperative with drafting. Recognising exploitation opportunities and sustaining reciprocity therefore emerge as distinct capabilities. These notes connect the results to repeated-game theory and develop future work on planning, relationship repair and richer interactions.

1 The question and the intervention

The experiment asks whether a short deliberation procedure helps a model respond to an opponent's incentives. Its two hosted models are Nemotron Nano and Super [6, 7]. The exact provider IDs, under the `nvidia/` namespace, are `NVIDIA-Nemotron-3-Nano-30B-A3B` and `nemotron-3-super-120b-a12b`.

Direct makes one thinking-off action request, capped at 128 completion tokens. *Draft* makes a thinking-on request capped at 1,024 tokens, then a separate thinking-off action request capped at 128 tokens [8]. Both receive completed game history. Only the current decision's draft is added to its action request. Each episode is assigned a sampling temperature $\tau \in \{0, 1\}$; all its request phases use that value. The nucleus-sampling setting is fixed at top-$p = 1$. Each model–procedure combination is an experimental *arm*.

Players maximise expected own cumulative points. Choices remain sealed until joint reveal. Neutral X/Y labels use both mappings and display seats. In canonical action labels, C means cooperate and D means defect:

Own action	Other cooperates	Other defects
Cooperate	$(3, 3)$	$(0, 5)$
Defect	$(5, 0)$	$(1, 1)$

Each pair lists (own points, other points). Write $R = 3$ for the mutual-cooperation reward, $S = 0$ for cooperating against defection, $T = 5$ for defecting against cooperation and $P = 1$ for mutual defection. These satisfy $T > R > P > S$.

2 Related literature and theoretical foundations

Rasmusen's *Games and Information* [1] supplies the organising ideas: strategies specify responses to histories, best responses maximise payoff against a given opponent, and equilibrium requires compatible best responses. Its treatment of finite and infinite repetition explains how the prospect of another encounter changes present incentives. Its discussions of incomplete information and the Axelrod tournament also motivate studying simple commitments alongside adaptive players. Here those ideas lead to two complementary tracks: exact payoff comparisons against disclosed policies, and observed interaction between models whose policies emerge during play.

Akata et al. [2] study finitely repeated games with language models, programmed strategies and human players. Their social chain-of-thought procedure encourages a model to anticipate the other player's behaviour and improves coordination in their experiments. This provides a close precedent for comparing decision procedures. Dilemma Dynamics uses a bounded draft with a separate final action request, making the procedure's token limits and request sequence explicit.

Fontana, Pierri and Aiello [3] examine Llama2, Llama3 and GPT3.5 in 100-round Prisoner's Dilemma interactions against random opponents with different cooperation probabilities. They check understanding of the rules and history and find substantial variation in responses to opponent hostility. Their work motivates our engineering checks and opponent-specific reporting. The present benchmark adds an exact own-payoff reference for each disclosed commitment, allowing cooperation and decision quality to be examined separately.

Pal et al. [4] infer strategies from decisions across possible recent histories, vary continuation probabilities, payoffs, horizons and framing, and assess equilibrium properties. Their approach emphasises the value of observing responses to histories that ordinary play may rarely visit. Our autonomous track records realised trajectories; its cooperation, breakdown and recovery statistics describe the interaction paths actually sampled.

Taylor and Bergen [5] study action–message incongruence in games with a communication phase. They vary the incentives and timing of messages and report associations between reasoning performance and misrepresentation. This places the present intervention in a wider behavioural context: payoff competence, cooperation and truthful communication are distinct properties of an agent. Our sealed-action design concentrates on the first two.

Together, these studies motivate a question about *when* deliberation helps: a change in average score may combine better exploitation of an unresponsive opponent, better preservation of reciprocity and different behaviour under uncertainty. The disclosed-policy track exposes those components, while matched temperature comparisons test their sensitivity to the sampling setting.

3 Repetition and the value of a future encounter

An *episode* is one complete interaction between two players. Round number is $t = 1, 2, \ldots$, and N denotes the total number of rounds in that episode. In a fixed-horizon episode, $N = H$, where the positive integer H is announced before play. In a geometric episode, continuation probability δ satisfies $0 \leq \delta < 1$. After each completed round, the game continues with probability δ and ends with probability $1 - \delta$, independently of the actions.

One equivalent construction draws an independent uniform variable $U_t \sim \text{Uniform}(0, 1)$ after round t, continuing precisely when

$$U_t < \delta.$$

The realised length then has distribution and mean

$$\mathbb{P}(N = n) = (1 - \delta)\delta^{n-1}, \qquad \mathbb{P}(N \geq t) = \delta^{t-1}, \qquad \mathbb{E}[N] = \frac{1}{1 - \delta}, \quad n, t \geq 1. \tag{1}$$

For $\delta = 0$, the game lasts one round. The first survival probability is one. Thus $\delta = .9$ produces ten expected rounds, with a hidden last round; an announced ten-round horizon gives players different information. The executable scheduler uses seeded, exact rational draws for the specified probabilities. Matched treatments reuse the same environment draws.

3.1 Committed responses

The disclosed opponents are AllC (always C), AllD (always D), Tit-for-Tat (C first, then the subject's preceding action) and Grim (C until a subject defection, then D forever). This makes the consequences of a current action computable. Rasmusen's distinction between a commitment and strategic equilibrium is useful here [1, Sections 4.1 and 5.1–5.2].

Against untriggered Grim, cooperation forever has value $R/(1 - \delta)$. Defecting now, followed by the best response to permanent punishment, has value $T + \delta P/(1 - \delta)$. Cooperation is weakly preferable exactly when

$$\frac{R}{1 - \delta} \geq T + \frac{\delta P}{1 - \delta} \quad \Longleftrightarrow \quad \delta \geq \frac{T - R}{T - P} = \frac{1}{2}. \tag{2}$$

All four payoff symbols were defined with the matrix; δ is the chance of another round. Equality allows both initial choices to be optimal.

Against initially cooperating Tit-for-Tat, three continuation patterns determine the optimal initial value: cooperate forever, defect forever, or alternate D,C. Their respective values are

$$\frac{3}{1 - \delta}, \qquad 5 + \frac{\delta}{1 - \delta}, \qquad \frac{5}{1 - \delta^2}.$$

The alternating pattern earns 5,0 repeatedly. Comparing all three shows that initial cooperation is optimal for $\delta \geq 2/3$. The tested probabilities .4, .6 and .9 therefore span different incentive regimes.

4 Exact payoff references and decision loss

Track A tests each disclosed policy at all three continuation probabilities. One-round AllC/AllD and announced ten-round Tit-for-Tat/Grim complete 16 conditions. For any condition, let s be the opponent's relevant memory state, $a \in \{\mathrm{C}, \mathrm{D}\}$ the subject's action, $r(s,a)$ the resulting own payoff and $f(s,a)$ the next opponent state. Let $V^*(s)$ be the maximum achievable expected total own payoff from state s; $Q^*(s,a)$ fixes the current action and allows optimal choices thereafter. In a geometric episode,

$$Q^*(s,a) = r(s,a) + \delta V^*(f(s,a)), \qquad V^*(s) = \max_{a \in \{\mathrm{C},\mathrm{D}\}} Q^*(s,a). \tag{3}$$

The implementation uses rational arithmetic. It certifies a candidate value function by checking $V^*(s) \geq r(s,a) + \delta V^*(f(s,a))$ for every state and action, with equality for at least one action in each state.

For a finite game, let h be rounds remaining, including the current one. Backward induction starts at $V_0^*(s) = 0$ and applies

$$Q_h^*(s,a) = r(s,a) + V_{h-1}^*(f(s,a)), \qquad V_h^*(s) = \max_a Q_h^*(s,a).$$

Against committed ten-round Tit-for-Tat or Grim, the optimal initial path is nine C moves followed by D, earning $9R+T = 32$ points. The standard finite-game backward-induction argument in which both players remain free to optimise uses a different opponent model [1, Section 5.1].

At a revealed decision, s_t is the pre-action opponent state and a_t the chosen action. Its nonnegative payoff gap is

$$g_t = V^*(s_t) - Q^*(s_t, a_t),$$

with the corresponding remaining-horizon values in a finite episode. Episode losses are

$$L_{\text{geometric}} = (1-\delta)\sum_{t=1}^{N} g_t, \qquad L_{\text{fixed}} = \frac{1}{H}\sum_{t=1}^{H} g_t. \tag{4}$$

Zero gap means that the current action preserves optimal achievable payoff. For example, C against AllC in a one-round game gives up $T - R = 2$ points; D has gap zero. Loss is expressed in normalised own-payoff points.

Equation (1) explains the geometric weighting: survival to round t already occurs with probability δ^{t-1}. The expectation of the realised sum in (4) therefore includes that survival factor. Applying it again to each recorded gap would count stopping twice.

Track B tests all ten unordered pairings of the four arms, including self-play, at $\delta = .9$. Here outcomes of interest are cooperation and joint return. For player $i \in \{1,2\}$, write $r_{i,t}$ for points in round t. A dyad's normalised joint return is $(1-\delta)\sum_{t=1}^{N}(r_{1,t} + r_{2,t})$. We report first-round mutual cooperation, cooperation per observed round and per episode, and breakdown/recovery counts with opportunity denominators. Long episodes receive more weight in a pooled round rate; rounds in one dyad share a history and are statistically dependent.

5 Replication and statistical comparisons

A block groups the related experimental conditions and arms. Each block has 64 Track A episodes (16 conditions for each of four arms) and ten Track B dyads per temperature. The pre-specified resource-based selection chose 48 blocks at each temperature: 6,144 benchmark episodes and 960 autonomous dyads, or 7,104 episodes in total. Four strata, defined by action-label mapping and display seat, each contain 12 blocks per temperature.

A fresh master seed fixes game schedules and order. Arms within a block share environment draws, and corresponding temperature episodes are adjacent in a frozen randomised order. This pairing controls stopping and presentation variation while spreading both temperatures through the same service window. The 48-block design was selected before scientific inference using measured token use and latency. Scientific outcomes did not enter that selection.

5.1 Primary drafting effects

Write $L_{m,k,\tau,b,c}$ for Track A loss in model $m \in \{\text{Nano}, \text{Super}\}$, procedure $k \in \{\text{Direct}, \text{Draft}\}$, temperature $\tau \in \{0, 1\}$, block b and condition $c \in \{1, \ldots, 16\}$. The block's condition mean is

$$\bar{L}_{m,k,\tau,b} = \frac{1}{16} \sum_{c=1}^{16} L_{m,k,\tau,b,c}.$$

For the contrast under consideration, let $\mathcal{B}_j$ be the set of blocks in stratum $j \in \{1, 2, 3, 4\}$ with complete observations for both procedures; $|\mathcal{B}_j|$ denotes its size. The drafting effect is

$$\Delta_{m,\tau} = \frac{1}{4} \sum_{j=1}^{4} \frac{1}{|\mathcal{B}_j|} \sum_{b \in \mathcal{B}_j} \left(\bar{L}_{m,\text{Draft},\tau,b} - \bar{L}_{m,\text{Direct},\tau,b} \right). \tag{5}$$

Negative values favour Draft. All conditions and strata have equal weight. The completed design has $|\mathcal{B}_j| = 12$ in every stratum. The point estimate requires at least one complete paired block in every stratum.

There are four primary contrasts, one for each model and temperature. Use 50,000 deterministic PCG64 bootstrap replicates, resampling whole blocks within strata. For family error level $\alpha = .05$ and $K = 4$ contrasts, each marginal interval has nominal coverage $1 - \alpha/K = .9875$. Its bootstrap quantiles are $\alpha/(2K) = .00625$ and $1 - \alpha/(2K) = .99375$. This Bonferroni allocation uses approximate bootstrap coverage. At least five complete paired blocks per stratum are required to construct an interval.

5.2 Temperature comparisons

Six secondary contrasts compare temperature 0 minus 1 for each of the four arms, and $\Delta_{m,0} - \Delta_{m,1}$ for each model. A positive value of the latter means a smaller drafting advantage at zero. Resampling uses the same matched block indices jointly across temperatures; secondary 95% intervals are nominal. All scheduled episodes completed, so every block contributes to the comparisons.

6 Evidence, execution and interpretation

The local CPU coordinates remote Nebius Token Factory inference. Inspect AI provides native samples, model events and custom scored outcomes [9]. Joint-reveal events carry hashes linking them to the canonical event chain. MLflow Skinny records run identities, parameters, metrics and artifacts in local SQLite [10]. NumPy supplies the bootstrap, PyArrow numeric Parquet exports and Matplotlib the comparison figures.

All 7,104 episodes completed: 3,552 at each temperature. The scientific run took 16.16 elapsed hours, from 24 October 2026, 14:10 UTC, to 25 October, 06:20 UTC, including a provider-overload interruption. Recovery preserved the first 3,406 completed episodes and their ledger prefix; the original and recovery source records are frozen separately. Saved successful responses retained their original identity throughout recovery.

Offline reconstruction produces episode histories, exact scores, decision/request tables, condition means, contrasts and checksums. Native Inspect coverage, canonical reconstruction and the final MLflow evidence heads were verified. Dependency locks, runtime records and source archives identify the executed implementation. Public action histories reproduce the scores and bootstrap comparisons without further model calls; private responses and draft text remain in the local evidence archive.

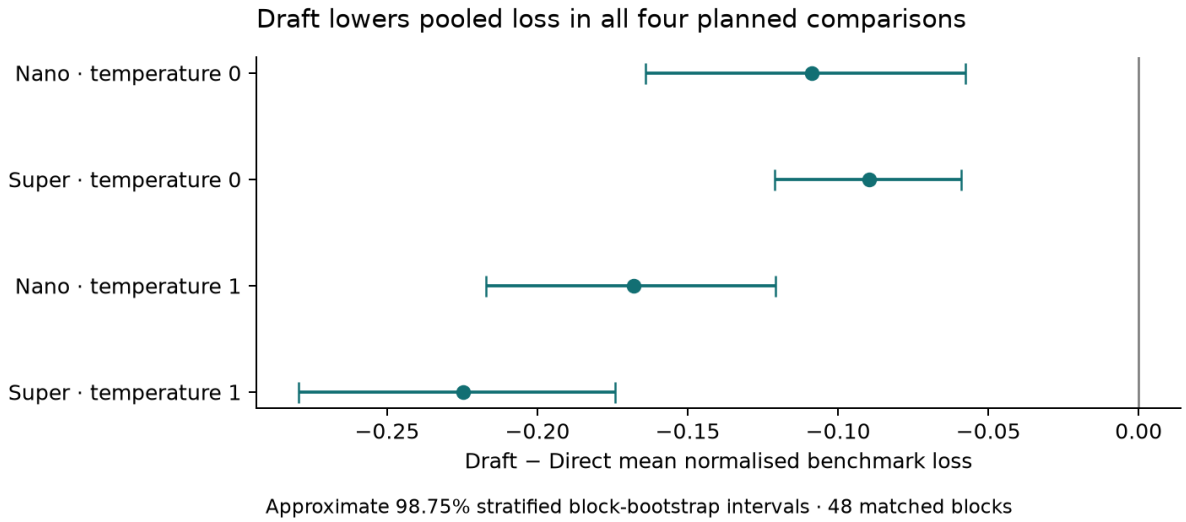
The design compares complete Direct and Draft procedures in the stated models, prompts and environments. Earlier development informed this design; the four primary contrasts were fixed before this campaign's scientific outcomes. Opponent, self-play and presentation diagnostics are exploratory. Primary intervals use approximate block-bootstrap coverage; secondary and exploratory intervals use their stated nominal levels. This context accompanies the results so that payoff competence and cooperative behaviour can be interpreted separately.

7 Results: better decisions against whom?

Each model–procedure arm has 768 benchmark episodes at each temperature. Table 1 gives the four planned primary comparisons; lower loss means less achievable expected own payoff forgone. The intervals use the pre-specified approximate four-comparison family treatment described above. Every interval lies below zero. Relative to each Direct mean, the reductions are 16.8%, 19.4%, 25.7% and 37.5%, respectively. These percentages describe loss reduction, not a percentage increase in earned reward.

Model	τ	Direct	Draft	Draft–Direct	98.75% interval
Nano	0	.6462	.5374	−.1088	[−.1640, −.0577]
Super	0	.4616	.3720	−.0896	[−.1212, −.0591]
Nano	1	.6541	.4861	−.1680	[−.2172, −.1209]
Super	1	.5988	.3741	−.2246	[−.2795, −.1742]

Table 1: Planned benchmark comparisons in normalised payoff points. All 48 matched blocks contribute, with 12 blocks in each presentation stratum.



7.1 The average hides a Tit-for-Tat reversal

Against Always Cooperate, drafting reduces loss by .5865 and .6042 for Nano at temperatures zero and one, and by .2094 and .5167 for Super. Against this unresponsive opponent, D is optimal at every round: an improvement therefore means more effective exploitation, rather than more cooperation. Super Draft also nearly eliminates loss against Always Defect. Nano Draft at temperature zero has a small opposite aggregate change against Always Defect (+.0068).

Against Tit-for-Tat the sign reverses for every model and temperature (Table 2). The reversal is concentrated in continuation .6, .9 and the announced ten-round game; at .4 drafting improves all four means. The .6 case rewards alternation, whereas .9 rewards sustained cooperation. Worse loss in both regimes indicates that a uniform instruction to cooperate more, or to defect more, would not explain all the errors. The 16-condition breakdown in Figure 1 shows how the drafting effect changes with the partner's response and the future value of interaction.

Opponent	Nano 0	Super 0	Nano 1	Super 1
Always Cooperate	−.5865	−.2094	−.6042	−.5167
Always Defect	+.0068	−.2083	−.1984	−.3385
Grim	−.0958	−.1511	−.1919	−.2151
Tit-for-Tat	+.2404	+.2103	+.3225	+.1717

Table 2: Exploratory Draft–Direct mean losses by opponent, averaging its four conditions equally (192 episodes per arm/opponent/temperature).

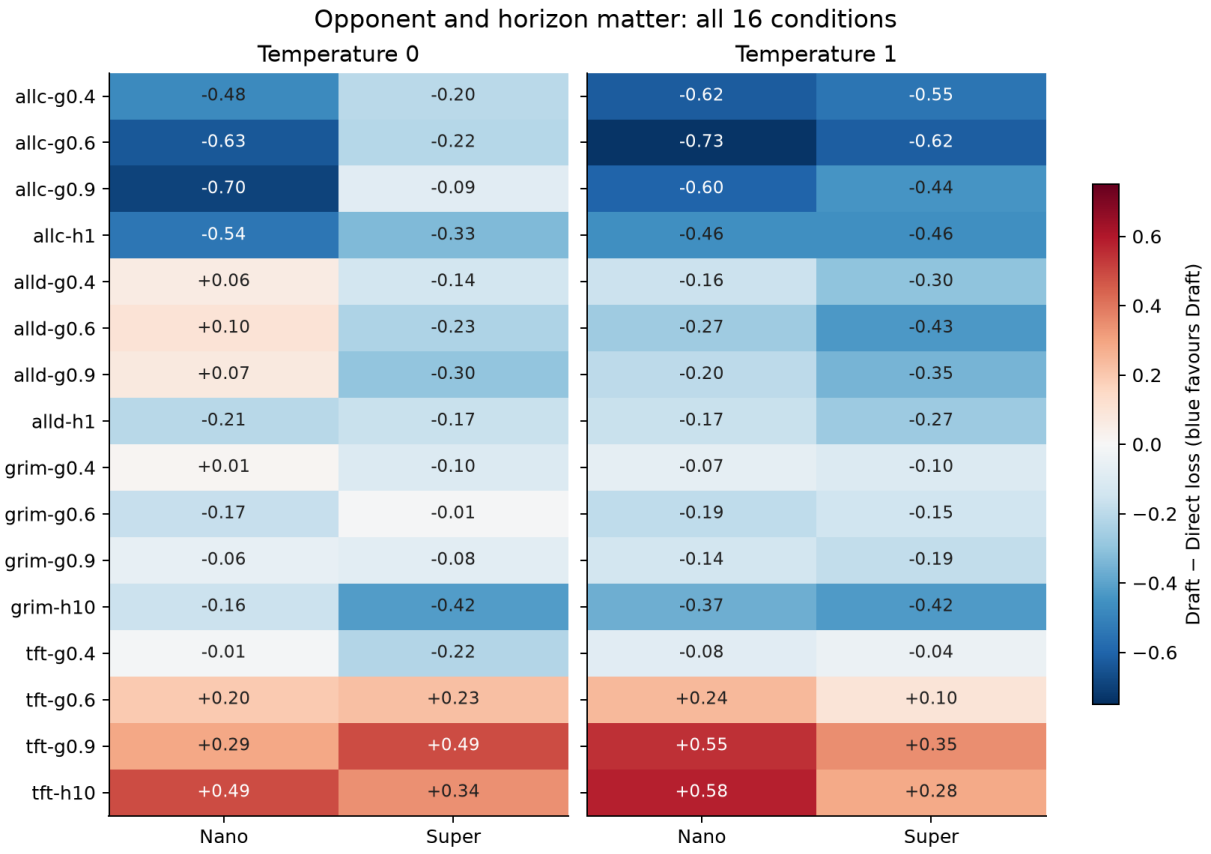


Figure 1: All condition effects, chosen for disclosure rather than significance. Blue favours Draft; red favours Direct. Each cell compares 48 episodes per arm. Condition codes g denote geometric continuation; h denotes announced length. These are exploratory point estimates; uncertainty is reported for the planned contrasts in Table 1.

7.2 Temperature changes the drafting advantage

The six planned secondary contrasts use nominal approximate 95% intervals. For Super Direct, temperature zero minus one is $-.1372$ $[-.1814, -.0942]$, while Super Draft changes by $-.0022$ $[-.0193, +.0147]$. Nano Direct changes by $-.0080$ $[-.0480, +.0316]$, and Nano Draft by $+.0513$ $[+.0199, +.0815]$. Thus zero temperature improves one Direct procedure markedly, worsens one Draft procedure, and leaves two contrasts near zero relative to their uncertainty. The Draft–Direct advantage narrows at zero for Nano by $+.0592$ $[+.0099, +.1084]$ and for Super by $+.1350$ $[+.0876, +.1849]$. These six intervals were not multiplicity adjusted. Temperature has no uniform preferred setting across the four arms.

7.3 Presentation sensitivity and finite planning

An exploratory presentation slice reveals a large Nano effect at temperature zero. When X means C, Draft–Direct loss is $-.4652$ in the first display seat and $-.2558$ in the second; when X means D, it is $+.1427$ and $+.1431$. Each stratum has 12 blocks. Seats show the same mapping-associated reversal, while Nano's temperature-one effects and all Super effects favour Draft in every stratum. Strata also have different stopping draws, so these estimates motivate a fully crossed label experiment rather than isolating a causal label effect. Equal weighting of all four strata is essential to the reported average.

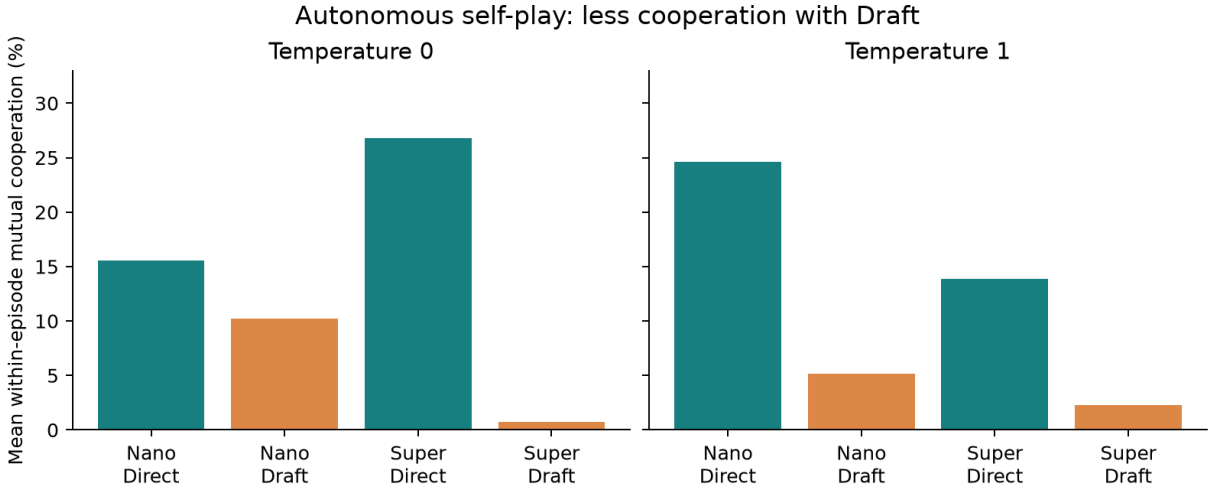
In the ten-round committed-policy games, the optimal own action path is nine C moves followed by D, with total payoff 32. Across all 768 such episodes, only one follows that path (Super Draft against Grim at temperature zero). Against TFT, Super Draft cooperates in round nine in 43/48 temperature-zero episodes and defects in round ten in all 48, yet never follows the complete optimal path. Its actions match a useful local endgame in many histories while losing payoff earlier. Against Grim at continuation .9, even Super Draft starts with the optimal C in only 26/48 episodes at zero and 17/48 at one. Together these patterns distinguish a useful local endgame choice from planning through the whole relationship.

8 Autonomous interaction: cooperation is a separate outcome

Every unordered pairing has 48 dyads and 376 realised rounds per temperature. Across all ten pairings, mutual cooperation occurs in 330/3,760 rounds (8.8%) at zero and 353/3,760 (9.4%) at one. The corresponding means of within-episode CC fractions are 8.9% and 9.8%. These two weighting schemes answer different questions; rounds in a trajectory are dependent.

Self-play arm	CC rounds, $\tau = 0$	J, $\tau = 0$	CC rounds, $\tau = 1$	J, $\tau = 1$
Nano Direct	61/376	2.7500	87/376	3.3729
Nano Draft	29/376	2.5083	15/376	2.3854
Super Direct	124/376	3.2063	58/376	3.1500
Super Draft	3/376	1.8667	9/376	2.0792

Table 3: Self-play cooperation and mean normalised joint return J. Draft lowers both point estimates for each model at each temperature.



For Super at zero, changing both self-play agents from Direct to Draft lowers mean normalised joint return by 1.3396 (exploratory nominal 95% paired-block interval $[-1.8375, -.9042]$). Super Draft self-play has 329/376 DD rounds, three first-round CC outcomes, and no return to CC in 325 opportunities. Super Direct self-play starts with zero first-round CC outcomes but later recovers CC in 82/218 eligible transitions. Its 124 CC rounds show that low initial cooperation can coexist with subsequent coordination.

The other self-play joint-return differences (Draft minus Direct) are $-.2417$ $[-.5458, +.0271]$ for Nano at zero, $-.9875$ $[-1.3063, -.6958]$ for Nano at one, and -1.0708 $[-1.4208, -.7521]$ for Super at one. These post-viewing comparisons change both players together and have no multiplicity correction. All ten pairings, including cross-model interactions, are released with CC/CD/DC/DD counts and breakdown/recovery opportunities.

The shared realised .9-continuation schedule averages 7.833 rounds, below its population expectation of ten. On this particular schedule, constant CC would yield mean $J = 4.7000$, and constant DD $J = 1.5667$. This helps interpret the normalised returns without mistaking a realised average for a population limit.

9 Resource observations

At each temperature every arm contributes 5,236 decisions across both tracks. Direct uses one successful request per decision and Draft two. Measured token use and request durations describe the resources used; usage-based cost estimates apply the rates recorded during the run. Within a model, Draft's estimated usage cost is about 10.0–11.9 times Direct's across the two tracks.

The thinking draft reaches its 1,024-token limit in 25.2% of Nano-zero requests, 41.0% of Nano-one, 37.1% of Super-zero and 56.1% of Super-one. Median request latencies range from .36–.69 seconds for Direct and .96–1.98 seconds across Draft's two phases. These are per-request observations; total decision delay also includes both phases, queueing and recovery. The resource profile makes the procedure concrete and motivates controlled comparisons of how computation is allocated between planning and action.

10 Future work

The study connects exact decision quality to the dynamics of a relationship. Its strongest contrast is that a procedure can improve average payoff choices while becoming less effective at sustaining reciprocity. The following directions extend that distinction into questions about how adaptive agents plan and interact.

10.1 From local choices to contingent plans

Tit-for-Tat changes the payoff-maximising pattern as continuation increases, while announced horizons create a specific endgame. These settings offer a way to study how agents represent a partner's state and carry a plan through time. A useful next experiment would compare explicit state-transition plans at several depths, with first actions and continuation choices scored separately. Matching compute across procedures and varying draft length independently would reveal which allocation of reasoning helps preserve valuable future responses. Fully crossing stopping seeds, action mappings and display seats would also show how robust those plans are to an equivalent presentation of the game.

10.2 Relationship formation, forgiveness and repair

Super Direct's return to mutual cooperation after uncooperative starts suggests that repair deserves its own target of study. Controlled partners could introduce known opportunities to forgive, test a cooperative move or restore trust after a fixed number of defections. Varying the frequency and timing of those opportunities would reveal when reciprocal exploration leads to a stable productive pattern. This would connect an agent's isolated choices to the path by which a pair enters, maintains or leaves cooperation.

10.3 Communication and noisy observations

Messages add another channel through which players can make commitments, coordinate expectations and explain apparent mistakes. Combining a communication phase with controlled action or observation noise would test how agents distinguish an accidental defection from a changed policy, and whether an explanation makes repair more likely. Comparing message content with subsequent actions would link payoff reasoning, cooperation and commitment reliability in a common protocol. The sealed-action game provides a clear baseline for that extension.

10.4 Longer interactions and changing environments

Longer horizons would allow repeated cycles of cooperation, breakdown and repair. Changing a partner's response rule during an interaction would test whether an agent updates its plan when familiar behaviour stops being informative. Separating history length from expected remaining interaction would help distinguish memory and context management from the incentive to preserve a valuable relationship. Repeated encounters across episodes could then study reputation and learning.

10.5 Diverse incentives and populations

Varying the payoff matrix would change the balance between temptation, joint surplus and punishment. Comparing symmetric and asymmetric roles, larger groups, and mixtures of simple policies, adaptive agents and human players would test how the observed capabilities transfer across social environments. Population-level interaction also raises questions about partner selection: an effective local policy may change which relationships persist and which strategies become common. These extensions would make it possible to connect exact small-game references to the broader coordination problems that motivate the study.

A Probabilities and the app's forecast inequality

The optional app protocol asks for an opponent-action forecast and two conditional future-payoff forecasts. These are elicited separately from the evaluation's action-only scientific requests. Their algebra helps explain the difference between immediate temptation and future incentives.

Let $\mathcal{H}_t$ be the completed history available before round t, and let $p_t \in [0,1]$ be the reported probability that the opponent cooperates in that round. Thus p_t concerns the opponent's *current action*; δ concerns whether a *later round exists*. The simultaneous protocol keeps the opponent's current choice sealed while the subject chooses.

Let $F_{a,t}$ be the reported expected own total from the next round onward, conditional on continuation and on choosing $a \in \{\mathrm{C}, \mathrm{D}\}$ now. It includes future opponent responses and subsequent stopping. Whenever no further round is possible, define $F_{a,t} = 0$. The probability of reaching the next round is

$$\rho_t = \begin{cases} \delta, & \text{geometric game,} \\ 1, & \text{fixed game with } t < H, \\ 0, & \text{fixed game with } t = H. \end{cases}$$

The conditional future totals have bounds

$$0 \le F_{a,t} \le \begin{cases} 5/(1-\delta), & \text{geometric game,} \\ 5(H-t), & \text{fixed game,} \end{cases}$$

because five is the maximum own payoff per round. In the geometric case, conditioning on the next round means the remaining expected length is again $1/(1-\delta)$.

Write $\widehat{Q}_t(a)$ for expected own payoff according to these submitted forecasts. With the payoff matrix in Section 1,

$$\begin{aligned} \widehat{Q}_t(\mathrm{C}) &= p_t R + (1-p_t)S + \rho_t F_{\mathrm{C},t} = 3p_t + \rho_t F_{\mathrm{C},t}, \\ \widehat{Q}_t(\mathrm{D}) &= p_t T + (1-p_t)P + \rho_t F_{\mathrm{D},t} = 1 + 4p_t + \rho_t F_{\mathrm{D},t}. \end{aligned}$$

Cooperation is weakly preferred according to the forecasts precisely when

$$\rho_t (F_{\mathrm{C},t} - F_{\mathrm{D},t}) \ge 1 + p_t. \tag{6}$$

The right side is the immediate gain from defection, between one and two points; the left side is cooperation's forecast future advantage, weighted by the chance of reaching that future. Equality means a tie. At an announced final round, $\rho_t = 0$, so the immediate incentive favours D for every $p_t \in [0,1]$.

This check measures consistency between an action and submitted forecasts. Forecast accuracy can then be studied against outcomes that were actually observed. A one-step counterfactual holds the other player's sealed current action fixed; a longer alternative history requires specifying how that opponent would respond.

B Recreational games: tactics on grids and lattices

Tic-tac-toe and Dots & Boxes are mathematical games played on discrete grids or lattices. Both have perfect information: the position is fully visible and moves are sequential. Their exact references illustrate tactical choice and search.

B.1 Tic-tac-toe

Let x denote a board state including the player to move, $A(x)$ its legal moves and $f(x, a)$ the state after move a. Let $G(x)$ be optimal game value from the current player's perspective. Terminal values are win $+1$, draw 0 and loss -1, according to that same perspective. Because each move passes the turn, a nonterminal state satisfies

$$G(x) = \max_{a \in A(x)} [-G(f(x, a))].$$

The game is small enough for exhaustive minimax. Legality, search value and the player's explanation can therefore be checked separately.

B.2 Dots & Boxes

Here x includes drawn edges, box ownership and the player to move. Write $g(x, a)$ for the number of boxes captured by drawing edge a, and $W(x)$ for optimal remaining-box difference from the current player's perspective. An edge that captures a box grants another turn. Thus

$$W(x) = \max_{a \in A(x)} \left[g(x, a) + \sigma(x, a) W(f(x, a)) \right], \qquad \sigma(x, a) = \begin{cases} +1, & g(x, a) > 0, \\ -1, & g(x, a) = 0. \end{cases}$$

At a terminal state, $W(x) = 0$, since no boxes remain. The difference in boxes already owned is added separately to obtain the final score difference. The positive sign preserves perspective after an extra turn; the negative sign switches it when the turn passes.

Exact edge search is used within explicit state and computation limits, with incomplete searches labelled unknown. Restricted long-chain/loop endgames also admit specialised references. A box sacrifice can preserve control and increase the eventual score; its value depends on the remaining position and the validity of that reference's assumptions.

Tic-tac-toe rewards a line of three while perfect play produces a draw. Dots & Boxes divides a fixed set of boxes, so a temporary sacrifice can be valuable when it preserves control of later chains. The app presents these rules and tactical examples in its Recreational games area.

References

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